Fiscal Year:	FY 2010	**Task Last Updated:**	FY 01/05/2010
PI Name:	Bungo, Michael W. M.D.		
Project Title:	Cardiac Atrophy and Diastolic Dysfunction During and After Long Duration Spaceflight: Functional Consequences for Orthostatic Intolerance, Exercise Capacity, and Risk of Cardiac Arrhythmias		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS Increment 19 (info from PI 2/2009) NOTE: End date corrected to 9/30/2012 per J. Dardano/JSC and PI; previous end date showed 1/10/2009 in error (1/2009 information)		
Key Personnel Changes/Previous PI:	PI: Benjamin D. Levine, M.D. ; Co-PI: Michael W. Bungo, M.D.		
COI Name (Institution):	Levine, Benjamin (UT - Southwestern)		
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Task Description:	Cardiac atrophy appears to develop during spaceflight or its ground-based analogues, leading to diastolic dysfunction and orthostatic hypotension. Such atrophy also may be a potential mechanism for the cardiac arrhythmias recently identified in some crew members after long duration exposure to microgravity aboard the Mir space station. Recent work has suggested that cardiac atrophy may be progressive, without a clear plateau over at least 12 weeks of bed rest, and thus may be a significant limiting factor for extended duration space missions. The purpose of this experiment is to quantify the extent, time course, and clinical significance of cardiac atrophy and identify its mechanisms. The functional consequences of this atrophy also will be determined for cardiac filling dynamics, orthostatic tolerance at 1G and fractional G (Mars and Moon) conditions, exercise tolerance, and arrhythmia susceptibility both in space on the ISS, and following return to earth. Three specific aims will be addressed: Objective 1a: To determine the magnitude of left and right ventricular atrophy associated with long duration space flight, and to relate this atrophy to measures of physical activity and cardiac work in-flight. Magnetic resonance imaging (MRI) will be performed pre- and post-flight as the most accurate means of measuring cardiac mass, including sufficient repetitions to document recovery. Objective 1b: To determine the time course and pattern of progression of cardiac atrophy in-flight using cardiac ultrasound. Objective 2: To determine the functional importance of cardiac atrophy for cardiac diastolic function and the regulation of stroke volume during gravitational transitions, by using innovative, non-invasive imaging techniques to measure the dynamic component of diastole including: a) Strain Rate and Tissue Doppler Imaging (TDI) to quantify ventricular wall motion with high temporal resolution; b) Color Doppler M-mode echocardiography to quantify intraventricular pressure gradients and assess the magnitude of diastolic suction; and c) MRI with myocardial tagging to quantify the rate of untwisting. Objective 3: To identify changes in ventricular conduction, depolarization and repolarization during and after long duration spaceflight, and relate these to changes in cardiac mass and morphology (presence of fibrosis from late gadolinium enhancement and/or fat accumulation from MR spectroscopy). Although the direct assessment of arrhythmia risk in otherwise healthy subjects is imprecise, the goal of this objective is to combine the best currently available non-invasive tests of arrhythmia risk with high negative predictive value to demonstrate convincingly the absence of risk for life threaten arrhythmias. These tests will include quantifying: 1) Arrhythmia burden; 2) Delayed conduction via the signal averaged electrocardiogram (SAECG); 3) Dispersion of refractoriness from QT dispersion (QTD), QT variability and T-loop morphology; and 4) Microvolt T-wave alternans (TWA) during daily activity including routine exercise. After completion of this study, the clinical expression of cardiac atrophy during long duration space flight will be defined clearly, and its significance for cardiac systolic and diastolic function at rest and during gravitational transitions will be elucidated. In addition, preliminary information will be obtained regarding ventricular conduction and repolarization that will provide either strong clinical reassurance, or pathophysiologic insight into the risk for cardiac arrhythmias. Once the magnitude, time course, and inciting factors for cardiac atrophy have been determined, effective countermeasures may be applied, focused on normalizing cardiac work and volume during long-duration spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained from these spaceflight experiments also will be relevant for patients after prolonged confinement to bed rest, as well as conditions that alter cardiac stiffness such as congestive heart failure, ischemic heart disease, and normal aging.
Task Progress:	Results: Data collection has just begun this past year with the launch of our first crewmember in June of 2009. We have successfully completed the preflight data collections on three crewmembers that are/were in flight this year as well as preflight data sessions on several backup crewmembers. In flight data collections to date have been successful and included the successful in-flight repair of the ISS echocardiogram in November 2009 – an equipment failure that would have had serious impact on a large portion of our flight data. Data collection around these non-nominal operations was abbreviated but should not have significant impact if the remainder of the study is accomplished as planned. Several in-flight data sessions fell outside the “windows” specified in the experiment document. One pre-flight MRI scan was unable to be obtained with the gadolinium delayed enhancement. Final impact of these anomalies will not be sufficiently defined until all data is collected and analyzed, since the impact of biologic variability versus temporal variability will not be defined until those data are available; however, at this time, it is not expected that these anomalies will have critical impact upon the ability to derive sufficient conclusions from data analyses. Based upon this most recent failure it is appropriate to emphasize the modular design of this experiment. This project was designed, from its inception, to address a number of the key science and operational questions that NASA requires for the continued progression of the manned space flight program. It was designed to use the highest fidelity equipment, the most sophisticated analysis techniques, and the methods that have been rigorously tested in the medical literature and universally accepted as providing high reliability. As the limited discussion presented above details, each technique contributes to several objects and each objective can be at least partially achieved by several of the techniques. This design allows for significant data “usability”. Thus, deviations from the initially proposed schedule and the initially proposed techniques will have impact upon the fidelity of conclusions that may result from the project, but the overall impact may be mitigated by the successful completion of other components of the study and protection against “catastrophic” failure ensured. Not all in-flight data has yet been down-linked. We have collected post flight data on our first two ISS returning crewmembers during the month of December 2009. We are hesitant to report too many details at this time for several reasons: (1) complete data sets are not available at the time of this report on any except one crewmember, (2) complete data analysis is being done in a “blinded fashion” thus reducing analysis bias and cannot be completed until more data sets are available, (3) our experiment calls for addition

	echocardiogram analysis at an independent core lab and these analyses will take additional time and effort, and (4) crew privacy needs to be assured when limited data is available that can be deduced back to the source. Having given those disclaimers, preliminary findings at this time suggest that in-flight cardiac work is reduced to approximately 25% of that seen before flight. In spite of this baseline unloading of the heart, crewmembers report that their exercise sessions in-flight are more regular and more sustained than their prior Earth bound experiences resulting in their perception that they exercise “more” in-flight. Cardiac mass as analyzed post-flight appears to be preserved, perhaps as a result of this sustained training effect. Detailed analyses of diastolic function are reserved for a future report. No malignant cardiac arrhythmias were noted, although additional predictive risk-analysis based upon the Holter data is still being compiled.
Bibliography Type:	Description: (Last Updated: 01/12/2021)
Significant Media Coverage	Bungo MW. "At the Heart of America’s Space Program: the Cardiovascular Effects of Weightlessness." University of Texas Medical School at Houston Cardiology Grand Rounds Presentation., April 2009., Apr-2009